



CLEAN FUEL CLEANER SKIES

Ethanol to Jet Fuel

Joshua Olusoji Omar Zaruk

AGENDA

01

Process Overview

Jet Fuel (SAF) production & importance

02

Process Safety & Environmental Analysis

Chemical & Operational Safety

03

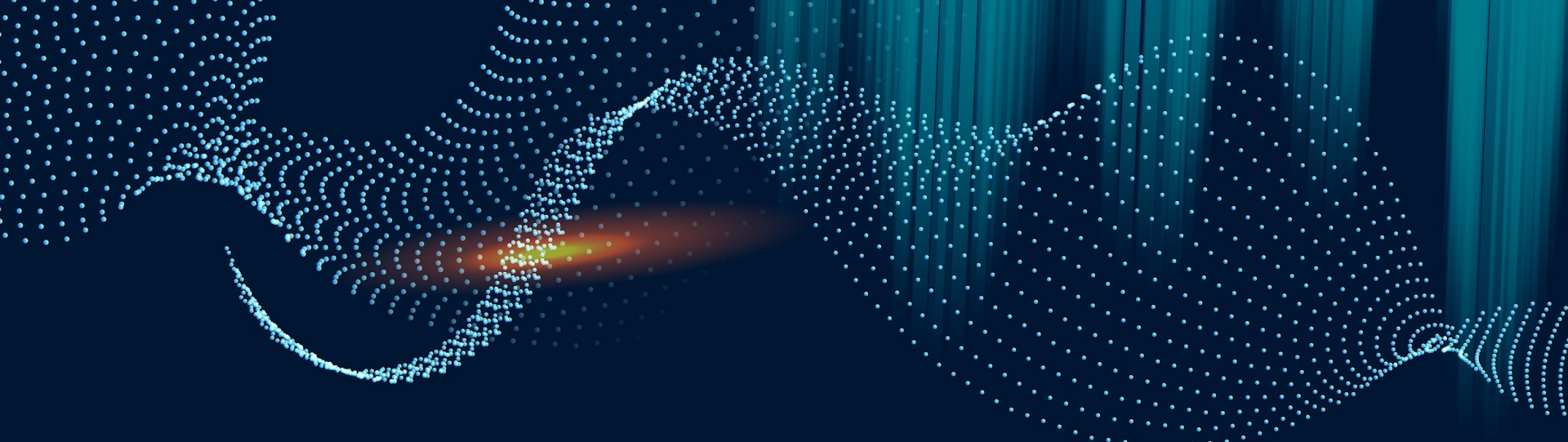
Process Controls Analysis

Process Control Loops and Strategy

04

Process Economics Analysis

CAPEX, OPEX, Income, NPV, and Sensitivity



01

Process Overview

Why SAF and How?

The Global Warming Problem

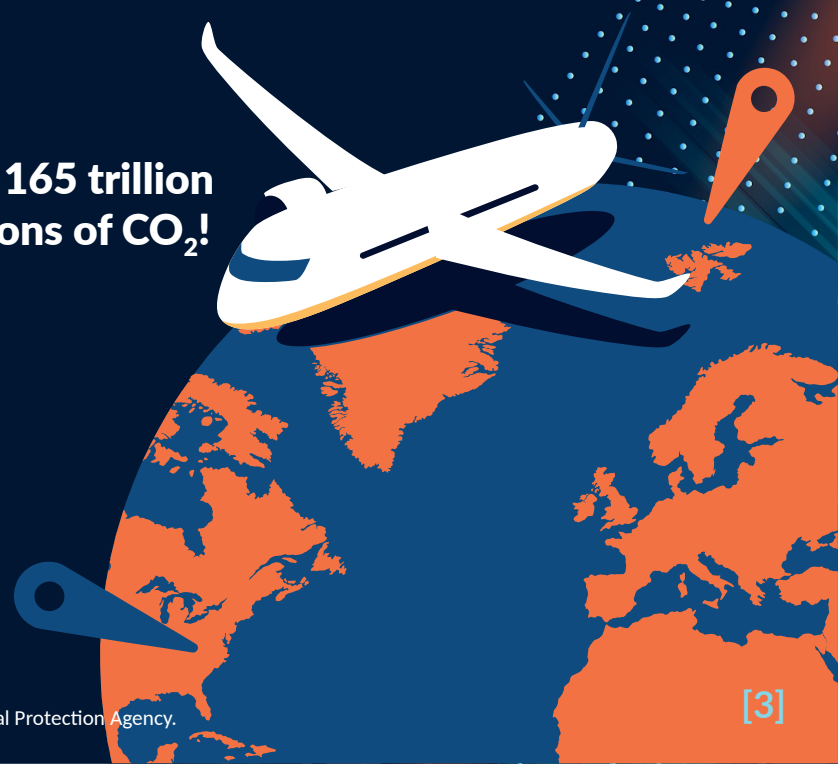
29%

Of all green house gas emissions are due to transportation

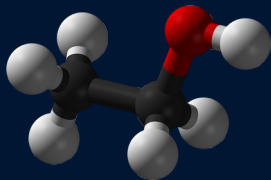
Approx. 165 trillion metric tons of CO₂!

9%

Of which is via aircraft!



Chosen Process



Ethanol



Jet Fuel

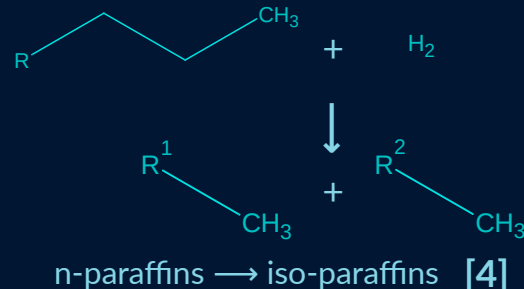
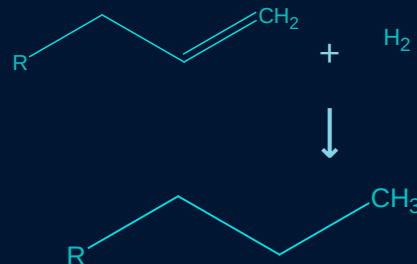
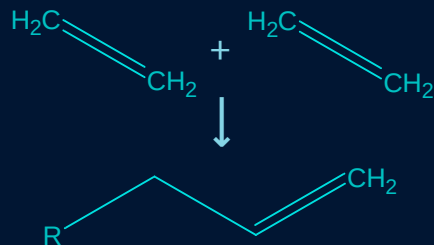
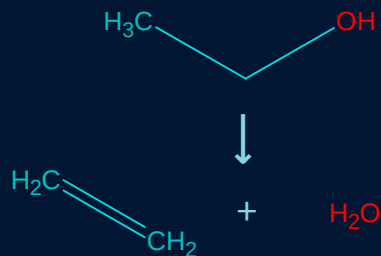


Dehydration

Oligomerization

Hydrogenation

Hydroprocessing



PLANT DESIGN BASIS

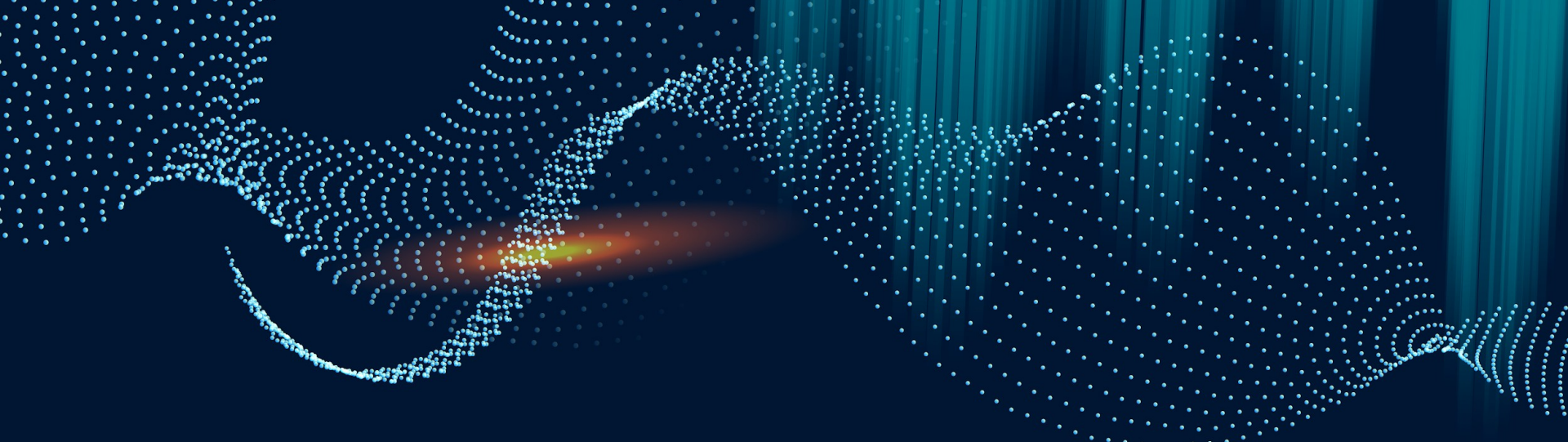


Location: Wilkinson County, Georgia

In-state feedstock & access to Port Savannah for international transactions!

Close proximity to Hartsfield-Jackson & Savannah Hilton airports!

Production: 10,000 Barrels/Day
This high daily production would **maximize profitability.**



02

Process Safety & Environmental Analysis

Plantwide Operational and Chemical Safety

Process Outputs, Emissions & Waste

Category	Representative Stream	Why It Matters	Typical Handling
Primary product	Jet Fuel Range Hydrocarbons	Main Economic Output	Product storage, QA/QC testing, controlled transfer
Byproduct	Naphtha Range Hydrocarbons	Side Economic Output	Condensation, reflux drum handling, product storage
Byproduct	Diesel Range Hydrocarbons	Side Economic Output	Bottoms withdrawal and storage
Air emission	Heater Flue Gas	CO ₂ , NO _x , SO _x , and CO emitted into air	Low-NOx burners and air permit compliance
Fugitive emission	Valve Stem, Pump Seal, and Flange Leaks	Potential VOC and flammable vapor release	Maintenance and equipment design
Liquid waste	Dehydration Process Water & Condensate Knockout	Oxygenates and dissolved organics require treatment	Wastewater treatment
Liquid waste	Cooling Tower/Boiler Blowdown	Contains salts and treatment chemicals	Industrial wastewater system
Liquid waste	Lubricants, Seal Oils, Cleaning Fluids Leaks	Inappropriate environmental releases	Spill kits
Solid waste	Catalysts & Adsorbents	Metals and contaminated solids require special disposal	Regeneration, metal recovery, return-to-vendor, or licensed disposal

Chemical Hazards

Material	Physical State	Primary Hazard	Main ETJ Concern
Ethanol	Liquid	Flammable Liquid	Pool fire, flash fire, ignition during handling, vapor release in hot service
Ethylene	Gas	Highly Flammable Gas	Explosion & flash fire risk; compression and purification service increase consequence
Hydrogen	Gas	Highly Flammable Gas	Very low ignition energy, leak detection challenge, high-pressure recycle gas service
Naphtha	Liquid & Vapor	Flammable Hydrocarbon	High volatility fluid in condenser, reflux drum, and storage
Jet & Diesel	Liquid	Combustible/Flammable Liquid	Hot liquid release, tank and pump fire exposure, product contamination
Oxygenate-rich condensate	Liquid	Flammable/Irritant Mixture	Potential VOC release and wastewater loading

Exposure Limits

Chemical	OSHA Limit/Note	NFPA Health	NFPA Flammability	NFPA Reactivity
Ethanol	PEL 1000 ppm	2	3	0
Hydrogen	Flammability more significant than toxicity	0	4	0
Ethylene	Asphyxiation & Flammability	1	4	2
Hydrocarbon	Exposure control and fire prevention	1	2-3	0
Alumina Catalyst	OSHA PEL 15 mg/m ³ dust, 5 mg/m ³ respirable	1	0	1

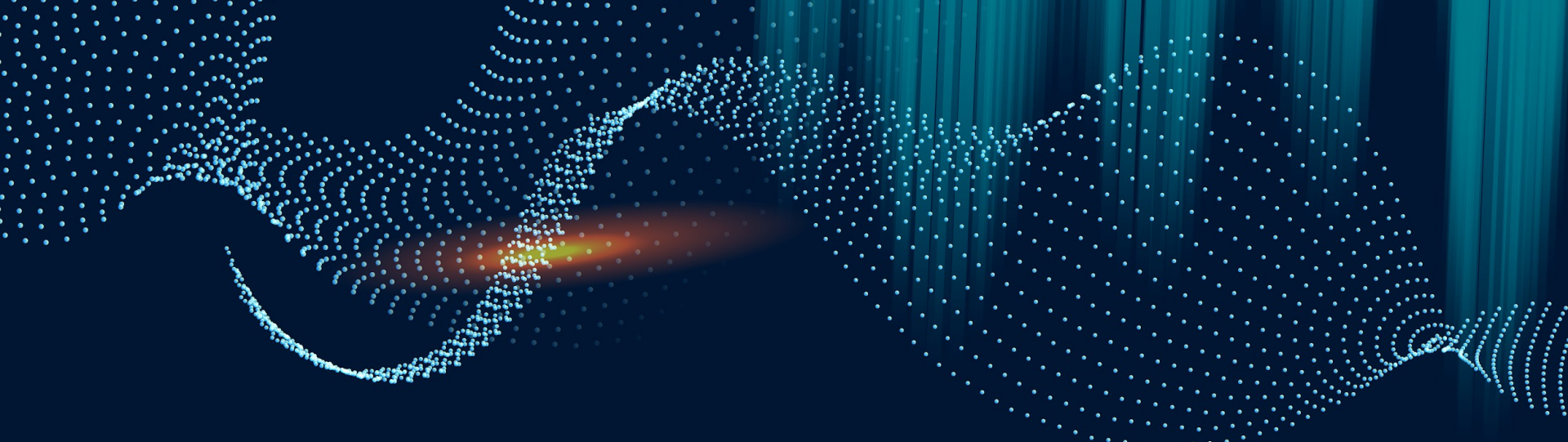
Chemical	OSHA Limit/Note	NFPA Health	NFPA Flammability	NFPA Reactivity
MTT Zeolite w/ Alumina Binder Catalyst	Catalyst dust, control exposure	1	0	0
Ni on Silica-Alumina Catalyst	Catalyst dust, irritation hazard	2	1	0
Pt on Alumina Catalyst	Catalyst dust, irritation hazard	2	0	0
Ni-W on FAU Zeolite Catalyst	Catalyst dust, irritation hazard	2	0	0
SAPO-34 w/ Alumina Binder Catalyst	Sieve dust, control exposure	1	0	0

Process-Wide Operational Safety

Hazard Type	Where It Appears	Potential Consequence	Primary Safeguards
Flash Fire	Ethanol, Naphtha, and Hydrocarbon storage/transfer	Personnel injury, product loss, equipment damage	Area classification, ignition source control, leak prevention, drainage, fire protection
Explosion / VCE	Ethylene compression and purification, Hydrogen recycle	Catastrophic overpressure, major equipment damage	Gas detection, ventilation, ESD, pressure protection, leak testing
High Temperature	Dehydration reactors, fired heaters, hot piping, reboilers	Burn hazard, thermal degradation, ignition of leaked material	Insulation, temperature control, guards, hot-work management
High Pressure	Compressors, Hydroprocessing loop, Fractionation	Rupture, jet release, escalation to fire	PSVs, pressure alarms, interlocks, design codes and inspection
Exotherm Control	Oligomerization and Hydroprocessing reactions	Runaway temperature, catalyst deactivation, off-spec product	Staged beds, heat removal, temperature alarms, feed cutback logic
Rotating Equipment	Pumps, Compressors, Fans	Seal leak, mechanical failure, caught-in hazards	Machine guarding, seal plans, vibration monitoring

Process-Wide Operational Safety

Hazard Type	Where It Appears	Potential Consequence	Primary Safeguards
Confined Space	Columns, Drums, Tanks, and Exchangers during maintenance	Asphyxiation, exposure, rescue complications	Permit-required entry, gas testing, isolation and LOTO
Working at Height	Tall columns, platforms, and overhead lines	Falls during inspection or maintenance	Handrails, fall arrest, access control, scaffolding procedures
Chemical Exposure	Caustic wash systems, wastewater handling, sample points	Eye/skin burns, inhalation irritation	PPE, eyewash/shower, procedures, and labeling
Loss of Utilities	Cooling water, steam, instrument air, power, lubricants, seal oils	Process upset, loss of control, or off-spec product	Utility alarms, fail-safe valves, PRV's, shutdown procedures, and backup power
High-Voltage Electrical Hazard	Motors, MCCs, Control Panels, Switchgear, Heaters, VFDs, lighting, and field wiring	Electric shock, arc flash, ignition of flammable vapor , equipment trip and fires	Area classification, grounding/bonding, arc-flash protection, LOTO, explosion-proof equipment, preventive maintenance
Moving Objects	Equipment like cranes and forklifts, falling tools during maintenance	Bodily harm from being struck on the head, arms and legs	Hard hats for small items, barriers and warning tape around elevated positions

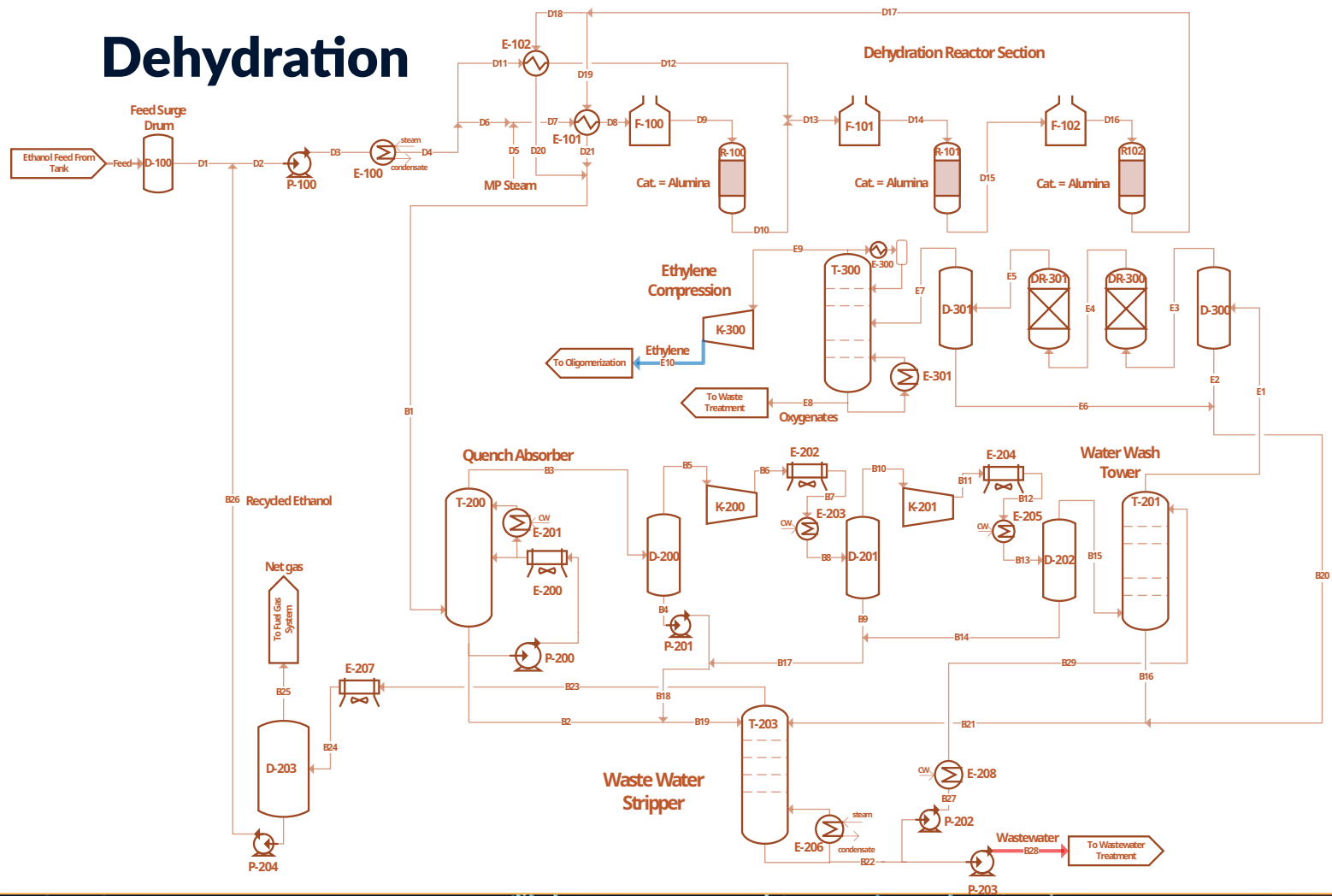


03

Process Controls

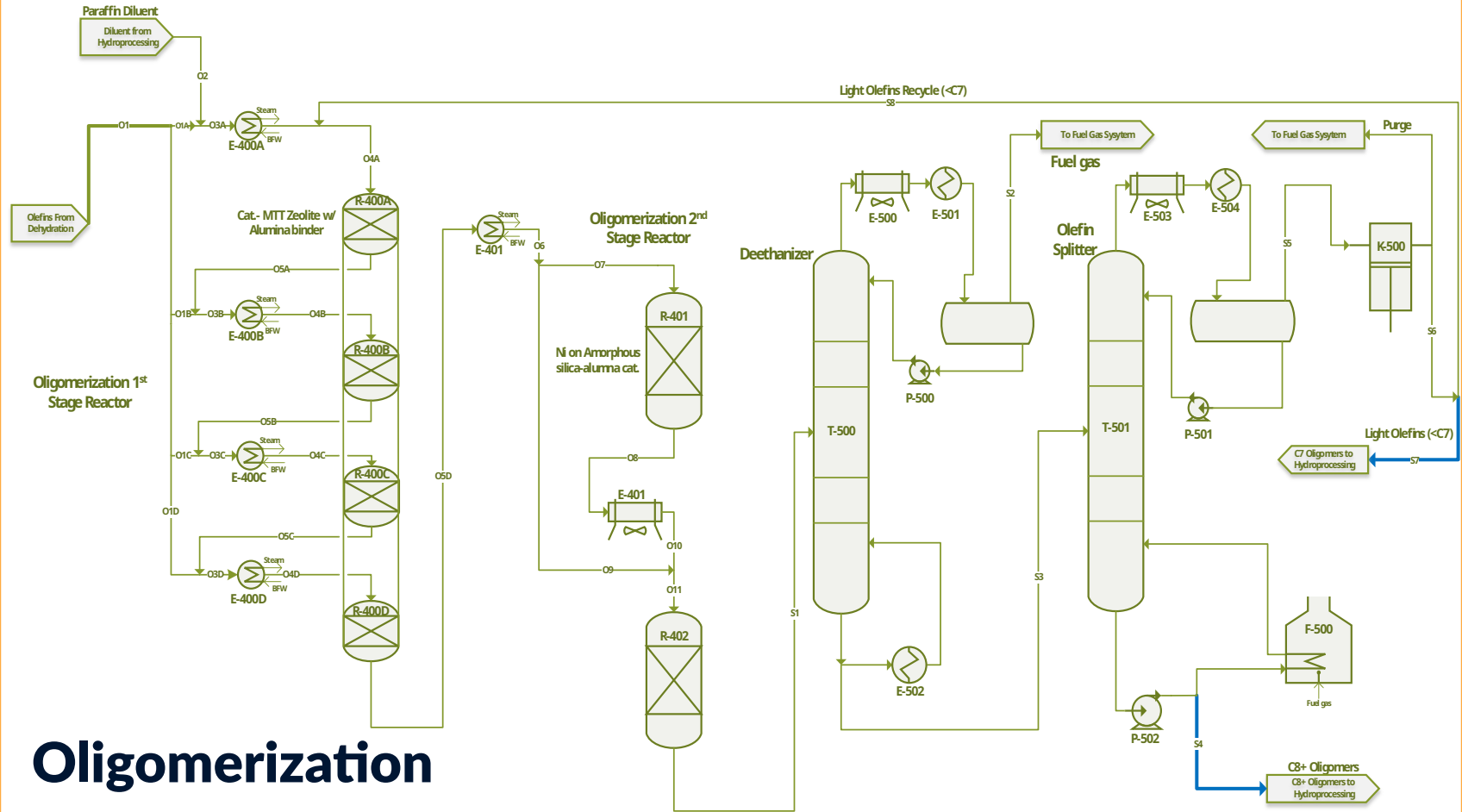
System Control Strategy and Control Loops

Dehydration



Dehydration Controls

Types of Equipment	Controls
Dehydration Reactors	Temperature, Pressure, Flow, Steam-Ethanol Ratio
Heat Exchangers	Temperature, Flow
Pumps	Pressure, Flow
Compressors	Pressure, Flow
Driers	Regeneration Temp, Moisture Content
Drums	Pressure, Level
Columns	Temperature, Pressure, Flows, Level

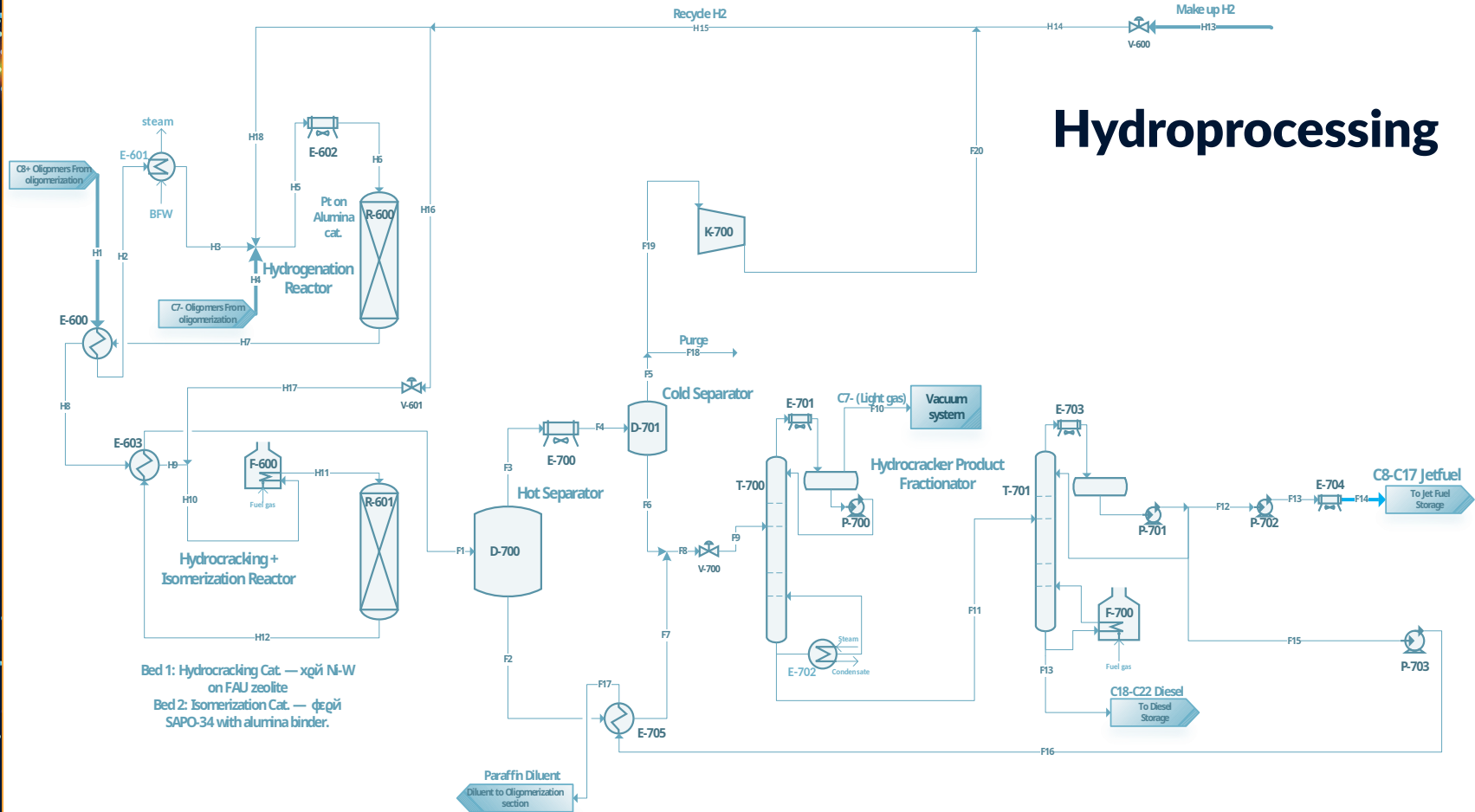


Oligomerization

Oligomerization Controls

Types of Equipment	Controls
Oligomerization Reactors	Temperature, Pressure, Flow, Paraffin-Olefin Ratio
Heat Exchangers	Temperature, Flow
Pumps	Pressure, Flow
Compressors	Pressure, Flow
Columns	Temperature, Pressure, Flows, Level

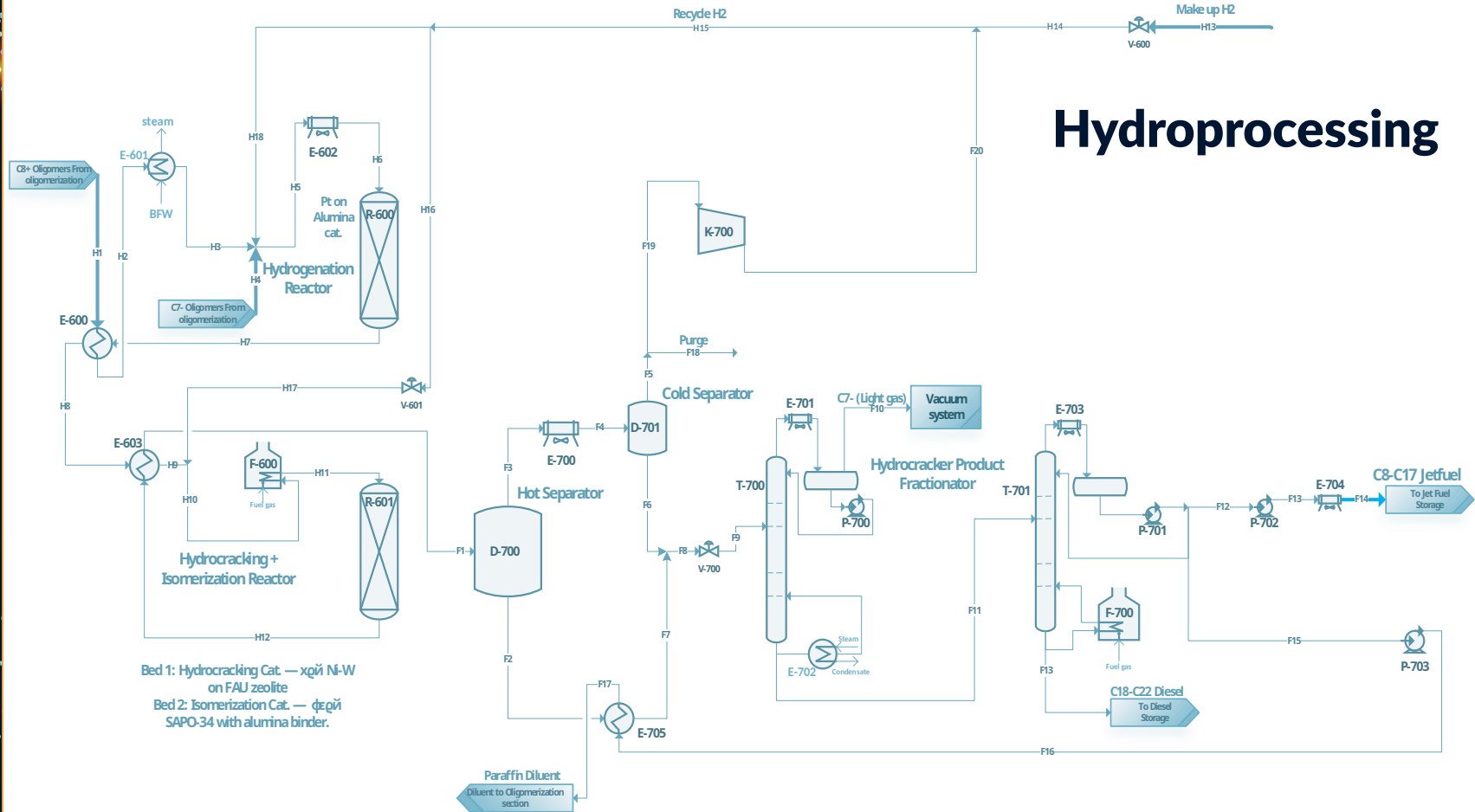
Hydroprocessing



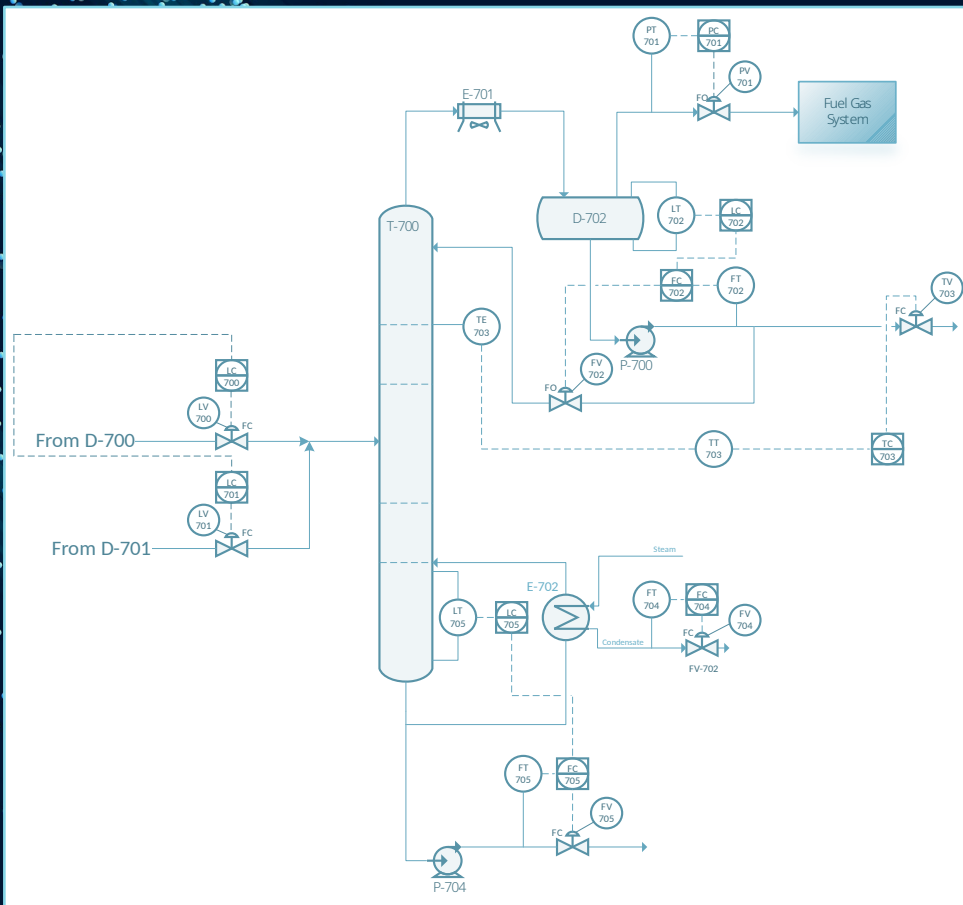
Hydroprocessing Controls

Types of Equipment	Controls
Hydrogenation & Hydrocracking Reactors	Temperature, Pressure, Flow, Hydrogen-Hydrocarbon Ratio
Heat Exchangers	Temperature, Flow
Pumps	Pressure, Flow
Compressors	Pressure, Flow
Drums	Pressure, Level
Columns	Temperature, Pressure, Flows, Level

Hydroprocessing



Fractionator Control Loop

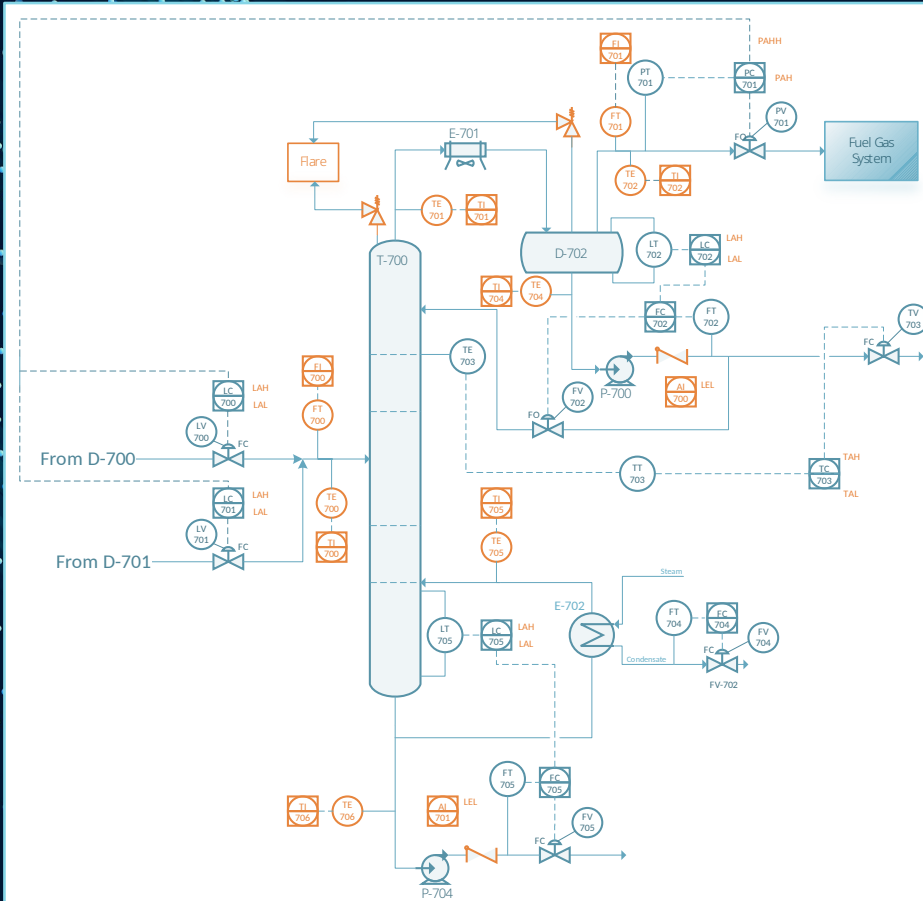


Manipulated Variable	Control Variable
Feed 1 Flowrate	D-700 Level
Feed 2 Flowrate	D-701 Level
Overhead Flowrate	Column Pressure
Reflux Flowrate	Accumulator Level
Distillate Flowrate	Column Temperature
Condensate Flowrate	Condensate Flowrate
Bottoms Flowrate	Column Sump Level

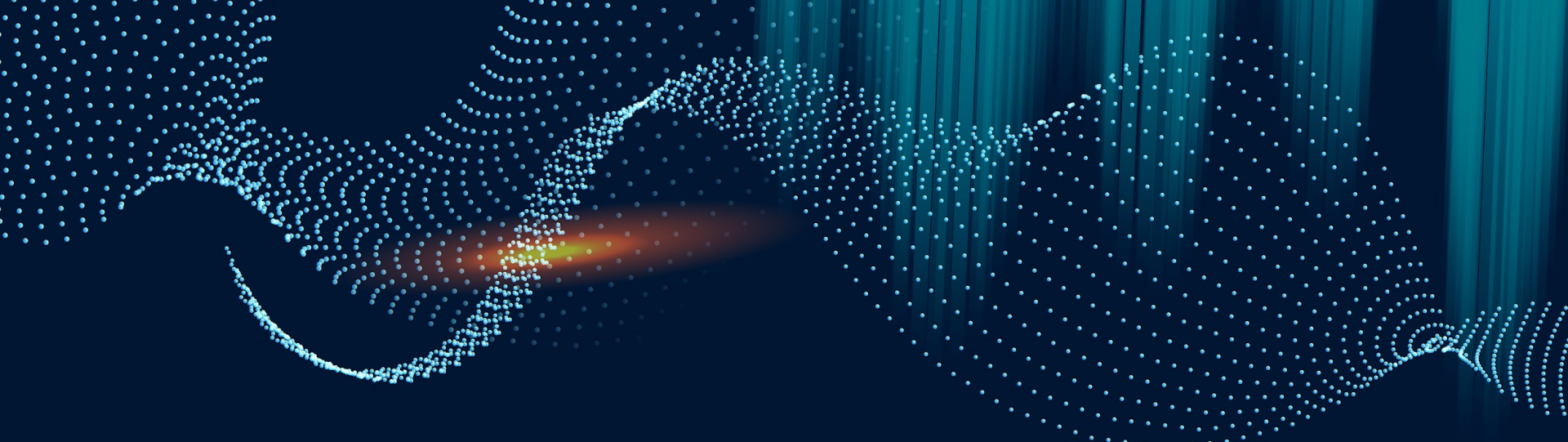
HAZOP Analysis

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Recommended Action
T-700 Feed	FV-700	Low Flow	Valve fails close, loss of level in D-700 or D-701, low pressure in drums	process upset, jet off spec	4	FI-700 will indicate low flow, naphtha spec allows some contamination from jet	2	4	
T-700 Feed			Likelihood						
			1	2	3	4	5		
T-700 Feed									Reactor section depressures into T-701, possible loss of containment and fire
									1
									FI-700 will indicate high flow, column pressure indication
									2
									1
									Provide PSV on column and overhead receiver, high pressure trip and low drum level trip to feed valves
T-700 Feed		1	1	1	2	3	NH		
		2	1	2	3	4	NH		
T-700 Feed		3	2	3	4	4	NH		
		4	3	4	4	4	NH		
T-700 Feed		5	NH	NH	NH	NH	NH		
T-700 Feed	FI-700	Loss of Containment	Flange leak, valve stem leak	venting hydrocarbon, possible fire or explosion	1	equipment design & maintenance	3	2	Install automatic ESD isolation on fractionator feed, improve valve/flange mechanical integrity with leak detection, LFI monitor
T-700 Feed	Outside of Section Scope	Variation in Feed Composition	Temperature excursion in R-601, catalyst deactivation, or temperature/pressure variation in D-700 or D-701	process upset, products off spec	3	temperature, pressure and flow measurements in R-601, regular sampling analysis	4	4	

Updated Control Loop



Node Descriptor	Recommendations
D-700 & D-701 Feed Drums	Level Control & Alarms
T-700 Feed	Flow & Temperature Indicator, Feed Valve Trip
T-700 Overhead	Flow & Temperature Indicator, PRV, Pressure Alarms
D-700 Overhead Accumulator	Level Alarms, PRV
T-700 Liquid Product	Temperature Indicator & Alarms, Check Valve, Analyzer
T-700 Bottom Product	Temperature Indicator, Level Alarms, Check Valve, Analyzer



04

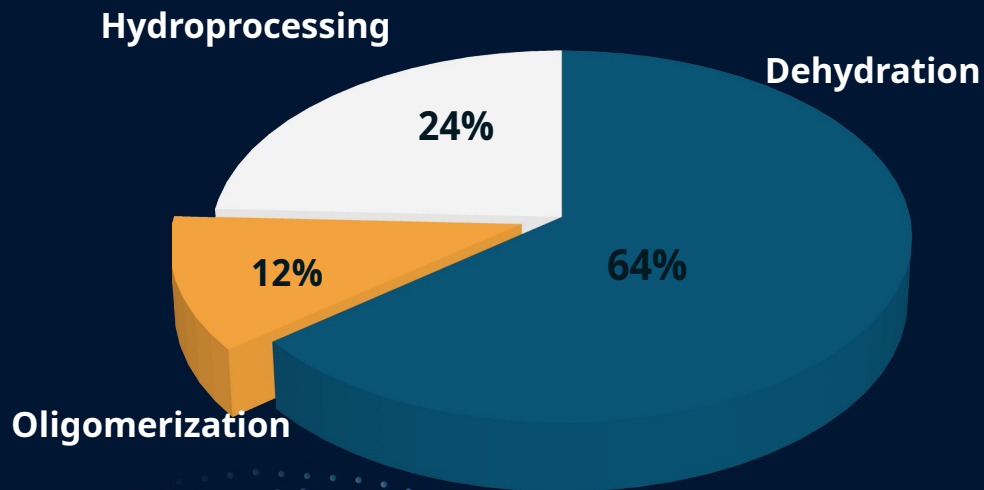
Process Economics

Cost and Value Analysis for the ETJ Process

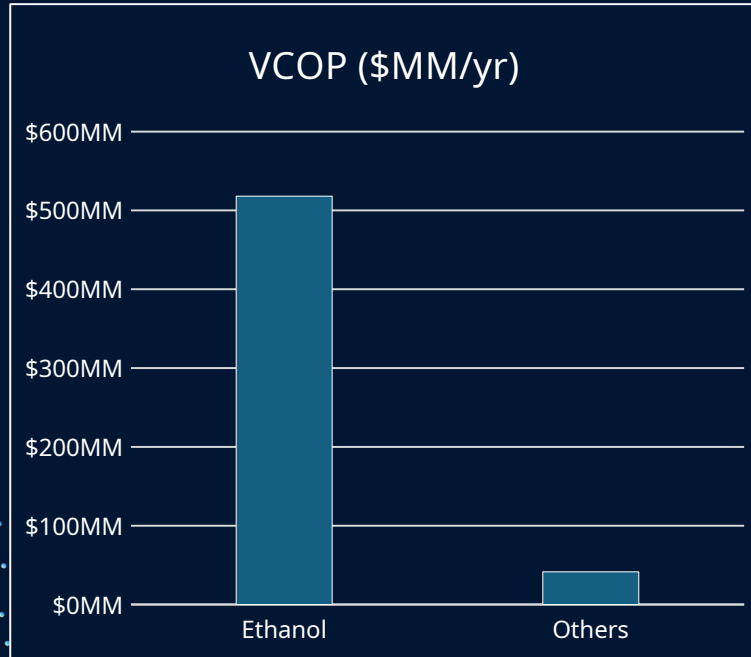
Capital Summary

Capital Cost Evaluation	
ISBL	\$95MM
OSBL	\$38MM
ECC	\$10MM
Contingency	\$10MM
WC	\$14MM
TCC	\$167MM

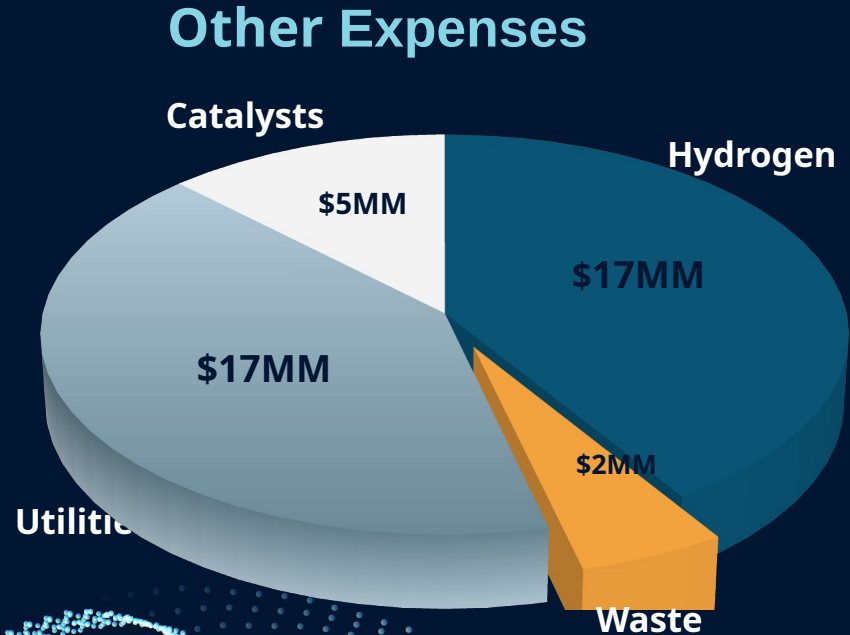
ISBL Distribution



Variable Cost of Production (VCOP)



Total VCOP: \$560MM/yr

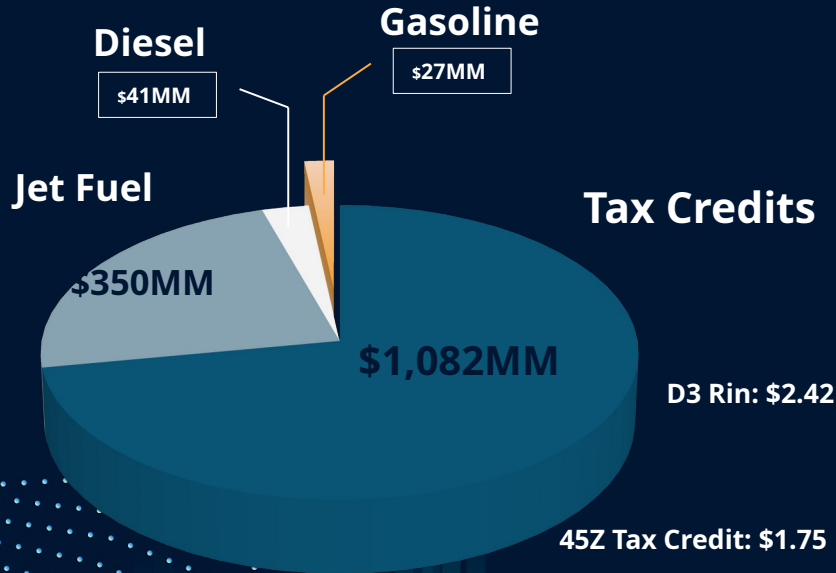


Fixed Cost of Production (FCOP)

Fixed Cost of Production Breakdown	
Number of Shifts	4
Operators per shift	5
Operator Salary (\$/yr)	\$70,000
Total Operator Compensation (\$MM/yr)	\$1MM
Supervision (\$MM/yr)	\$0.3MM
Direct Salary Overhead (\$MM/yr)	\$0.6MM
Maintenance (\$MM/yr)	\$5MM
Overhead Expenses (Plant, Taxes, Insurance, Land) , (MM/yr)	\$3MM
Total FCOP, (\$MM/yr)	\$10MM

Income Breakdown

Revenue Streams



Revenue: **\$1,500MM/yr**

Gross Profit: **\$930MM/yr**

Net Profit: **\$687MM/yr**

D3 Rin: \$2.42

45Z Tax Credit: \$1.75

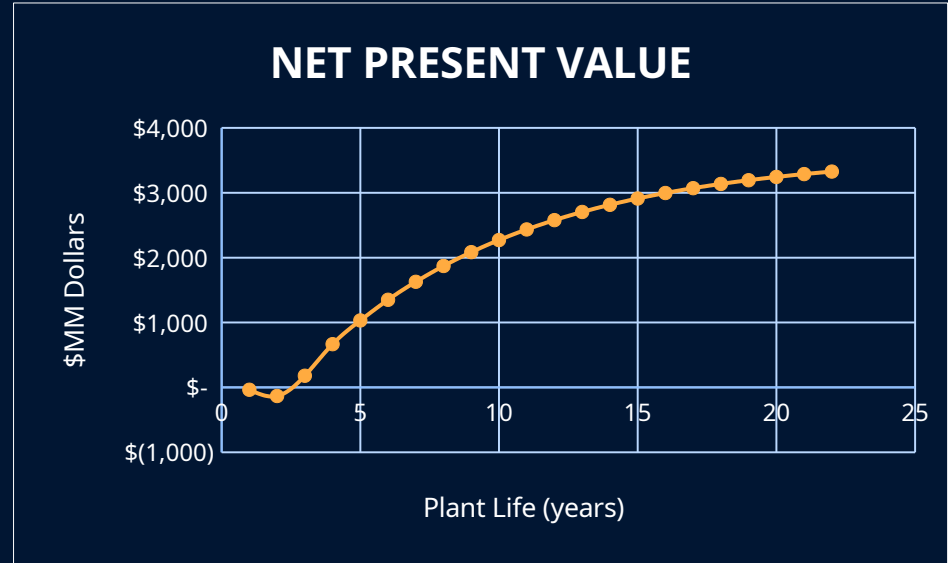
26% Net Tax Rate

Economic Evaluation Metrics

1.69yrs
SPBP

219%
DCFROR

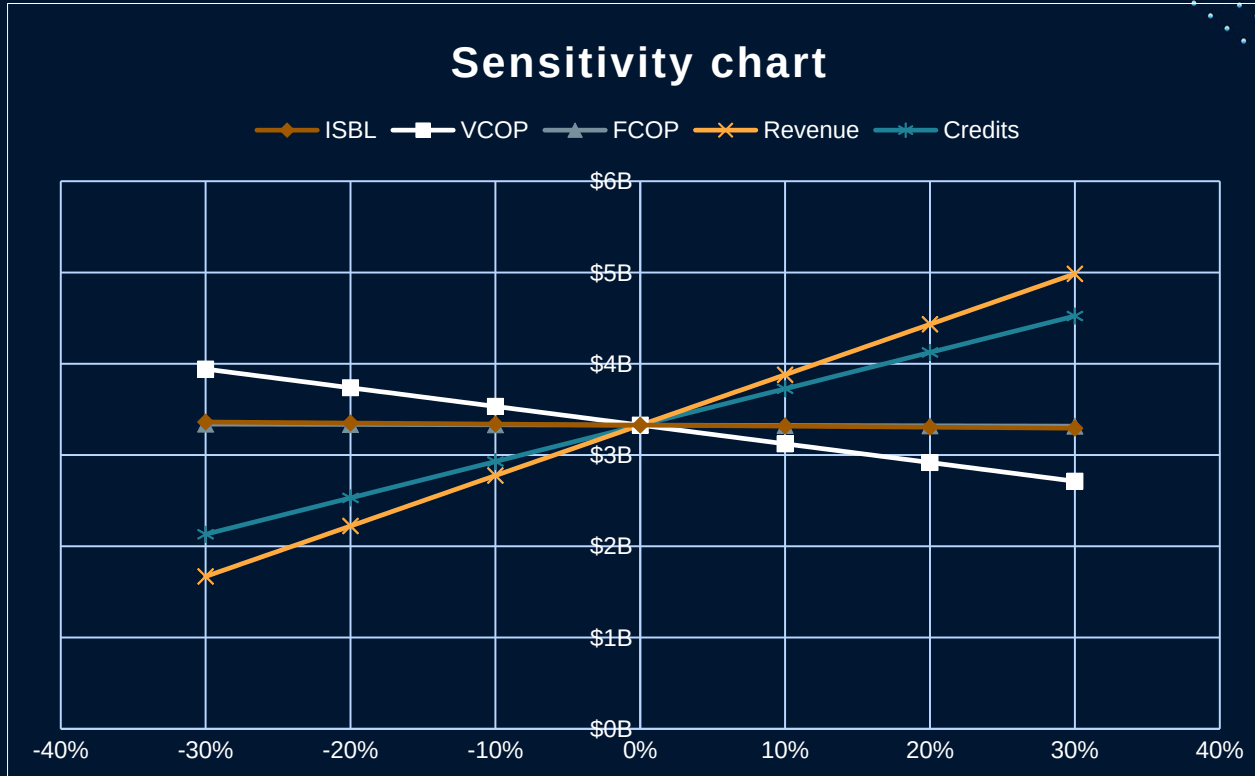
415%
ROI



\$3,313MM

NPV at Year 20

Sensitivity Analysis



Biggest Factors:
Revenue & Credits

Smallest Factors:
FCOP & ISBL

SPECIAL THANKS

**LANCE
BAIRD**

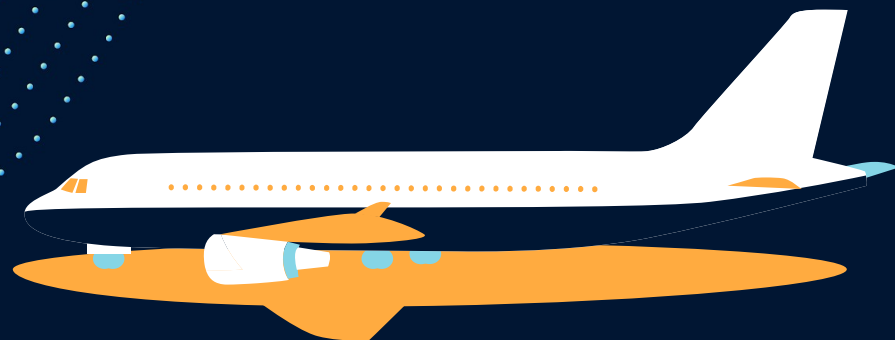
Mentor



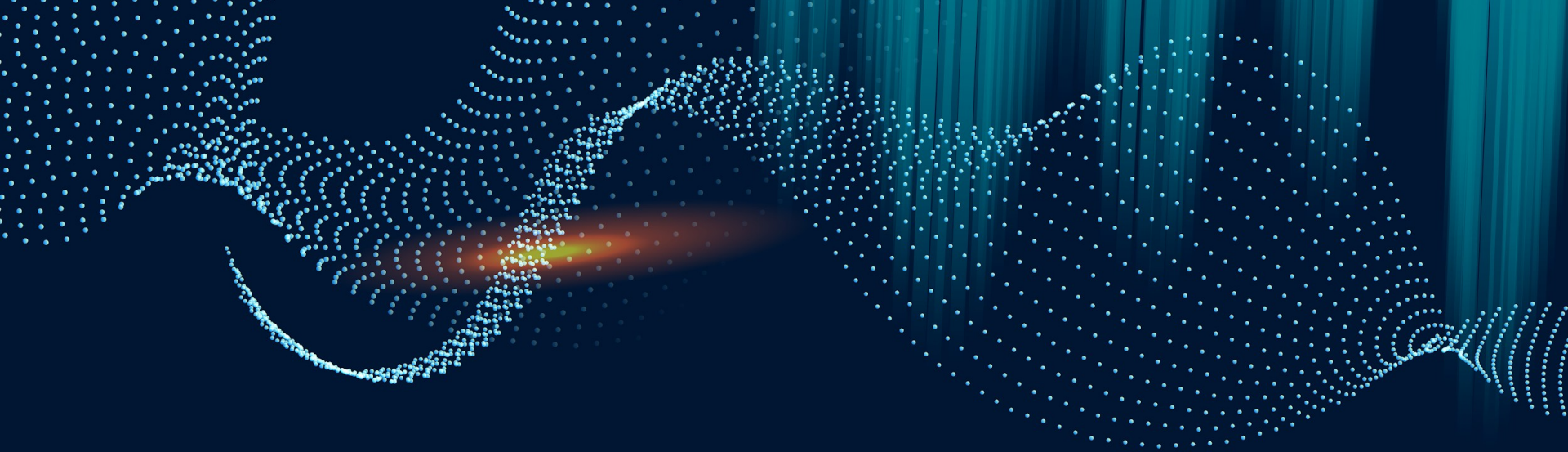
**BETÜL
BILGIN**
Advisor

References

1. Occupational Safety and Health Administration (OSHA), 29 CFR 1910.119, Process Safety Management of Highly Hazardous Chemicals.
2. U.S. Department of Energy, Sustainable Aviation Fuel Grand Challenge, official program overview and goals.
3. U.S. Environmental Protection Agency (EPA), National Emission Standards for Hazardous Air Pollutants (NESHAP) program materials for chemical and petroleum sector emission controls.
4. Ulrich, G. D., & Vasudevan, P. T. (2006). How to estimate utility costs. *Chemical Engineering*, 113(4), 66–69.
5. United States Environmental Protection Agency. (n.d.). *Sources of greenhouse gas emissions*.
6. U.S. Department of Energy. (2024). *Sustainable aviation fuel grand challenge: Tracking metrics and mid-2024 dashboard*.
7. U.S. Energy Information Administration. (2026). *Kerosene-type jet fuel prices: U.S. Gulf Coast (DJFUELUSGULF)*. Federal Reserve Bank of St. Louis, FRED



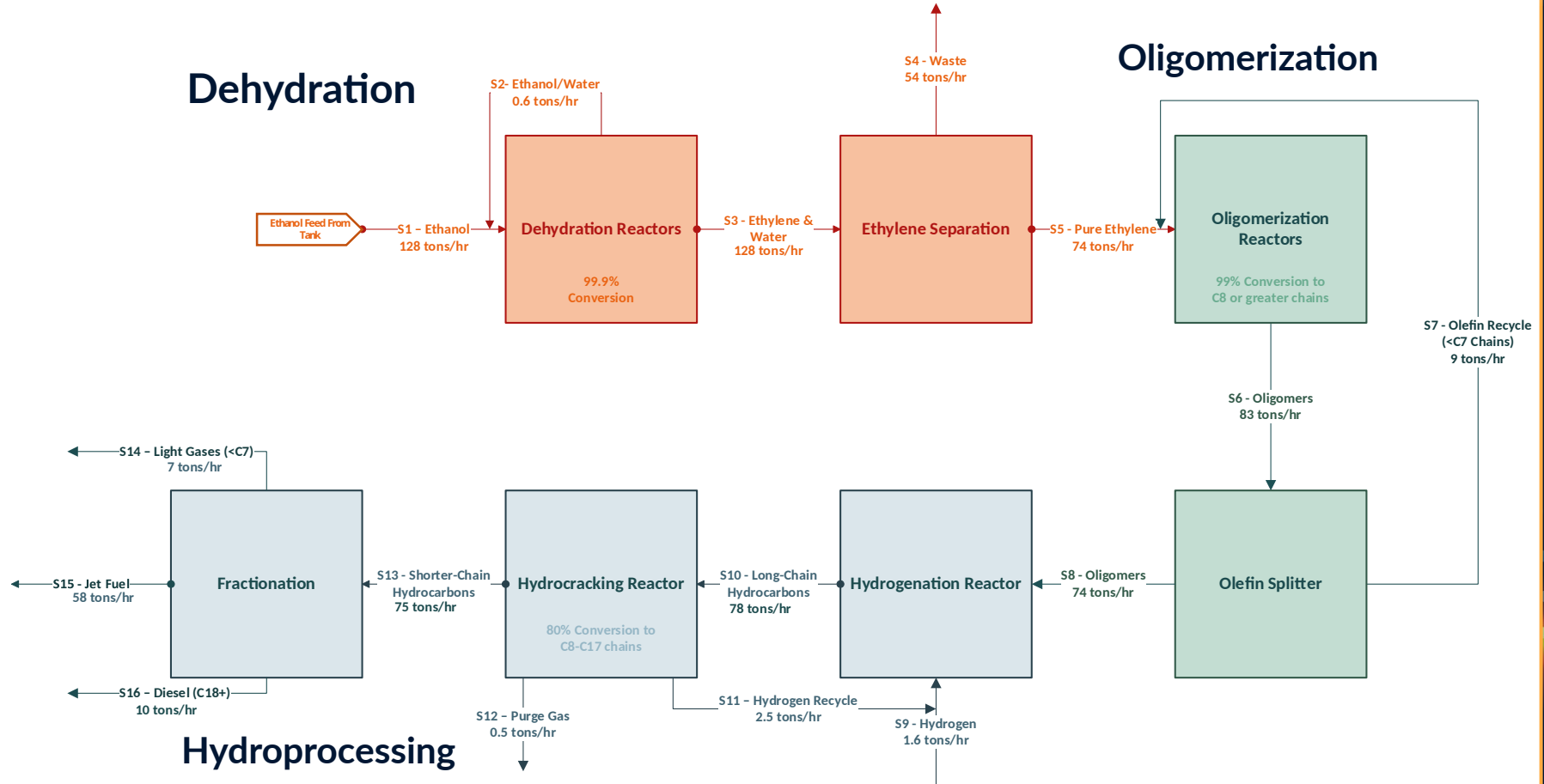
ANY QUESTIONS?



05

Additional Material

Block Flow Diagram & Material Balances



Utilities

Name	Fluid	Rate	Rate Units	Cost per Hour	Cost per year	Cost Units
Electricity	N/A	6,131	KW	367.85	\$2,942,813	USD/H
Cooling Water	Water	81,945,270	BTU/H	30.57	\$244,525	USD/H
Fuel Gas	Steam	282,696,200	BTU/H	1,266.48	\$10,114,763	USD/H
HP Steam	Steam	11,155,360	BTU/H	29.45	\$235,601	USD/H
LP Steam	Steam	4,577,057	BTU/H	9.15	\$73,233	USD/H
HP Steam	Steam	179,717,600	BTU/H	416.94	\$3,335,559	USD/H
Refrigerant	Propane	10,119,320	BTU/H	29.244835	\$233,959	USD/H

Total Utilities: \$17.2MM/yr

GHG Emissions For LanzaTech Ethanol

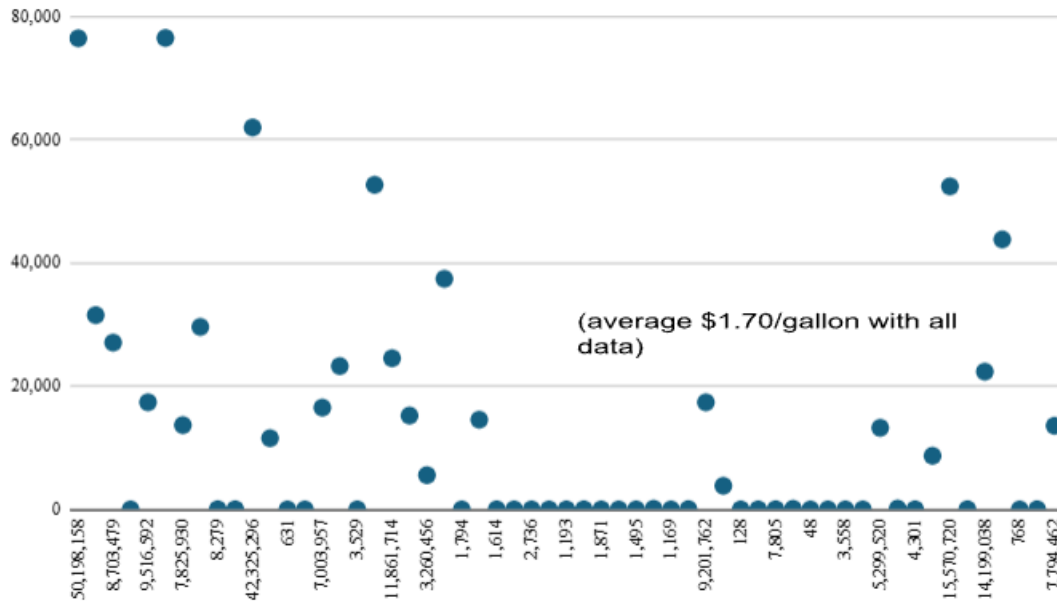
Table 3. GHG Emissions for LanzaTech Scenarios

scenario	GHG emissions (g CO _{2eq} /MJ ethanol)			
	BOF gas	corn stover	switchgrass	forest residue
net gas used ^a	-79.0	-80.2	-80.2	-80.2
feedstock procurement	0.0	11.2	14.9	6.6
utilities (heat and power) ^b	28.8	-6.3	-6.3	-8.1
other inputs ^c	1.8	1.3	1.3	1.3
anaerobic digester emissions	6.8	7.5	7.5	7.5
waste treatment ^d	0.9	1.1	1.1	1.0
ethanol transport	0.7	0.7	0.7	0.7
ethanol combustion	71.4	71.4	71.4	71.4
net EtOH GHG emissions	31.4	8.0	11.7	1.5
% reduction ^e	67%	92%	88%	98%

Retrieved from "Life Cycle Assessments of Ethanol Production via Gas Fermentation: Anticipated Greenhouse Gas Emissions for Cellulosic and Waste Gas Feedstocks."

Ethanol Price

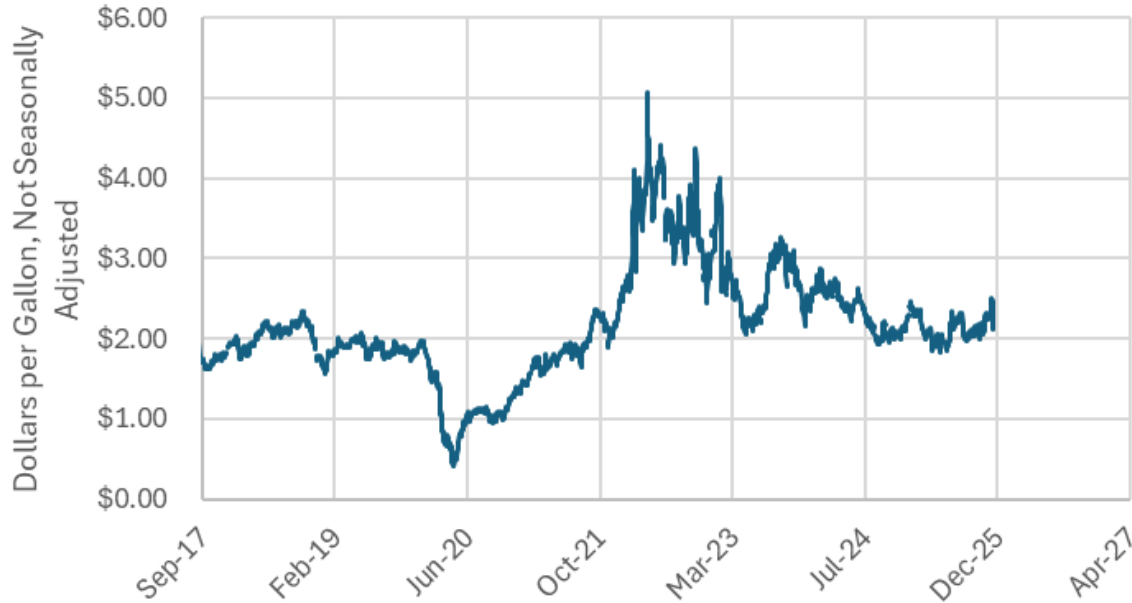
Value, US\$000 vs Imported gallons



Data Source: U.S. Census Bureau Trade Data

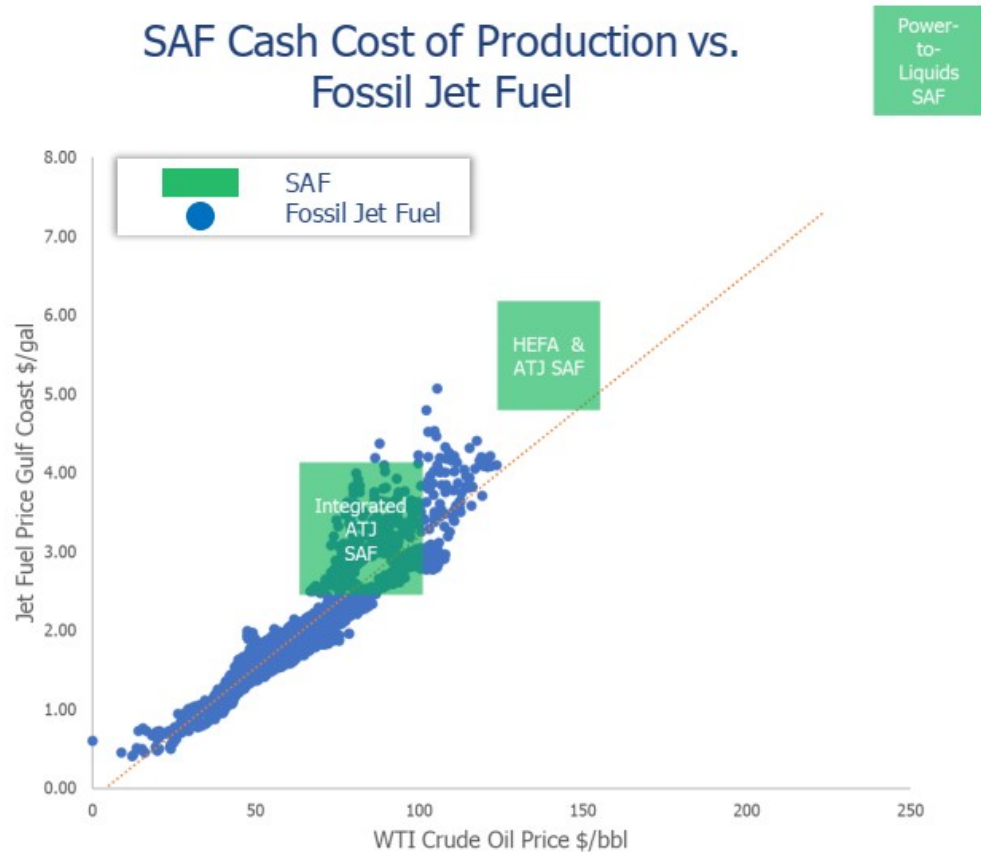
Jet Fuel Price

Kerosene-Type Jet Fuel Prices: U.S. Gulf Coast



U.S. Energy Information Administration, Kerosene-Type Jet Fuel Prices: U.S. Gulf Coast [DJFUELUSGULF], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/DJFUELUSGULF>, March 15, 2026.

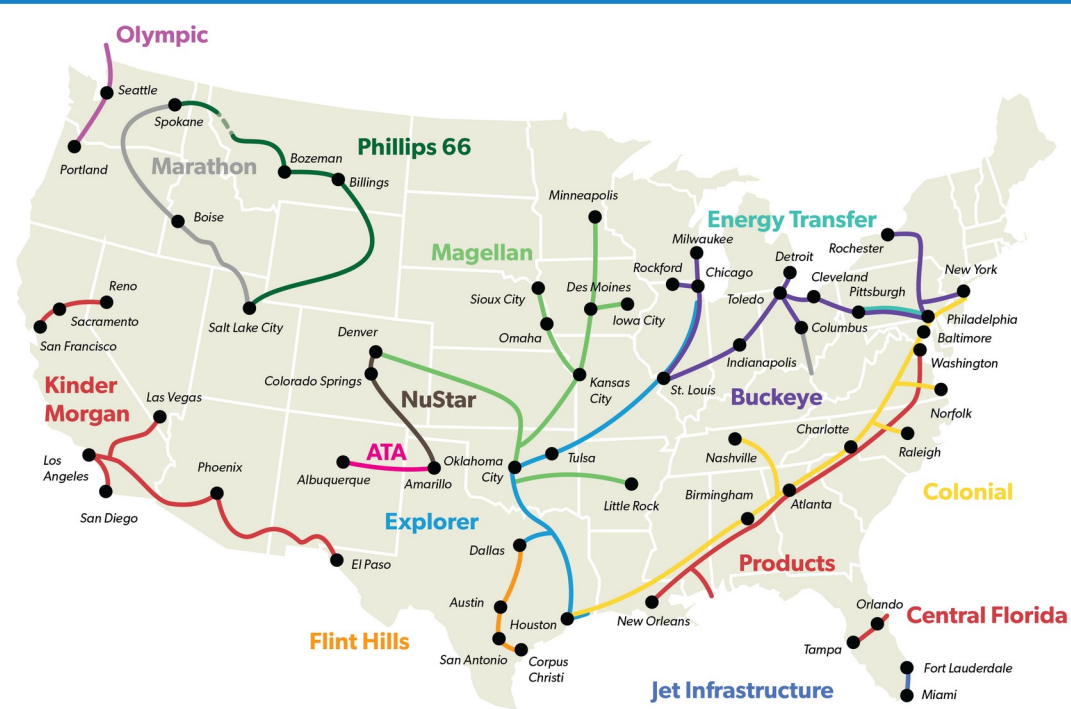
SAF vs. Fossil



Jet Fuel Pipelines

MAJOR U.S. PIPELINES CARRYING JET FUEL

Most jet fuel is supplied to U.S. airports through a combination of multiproduct pipelines and dedicated local pipelines.

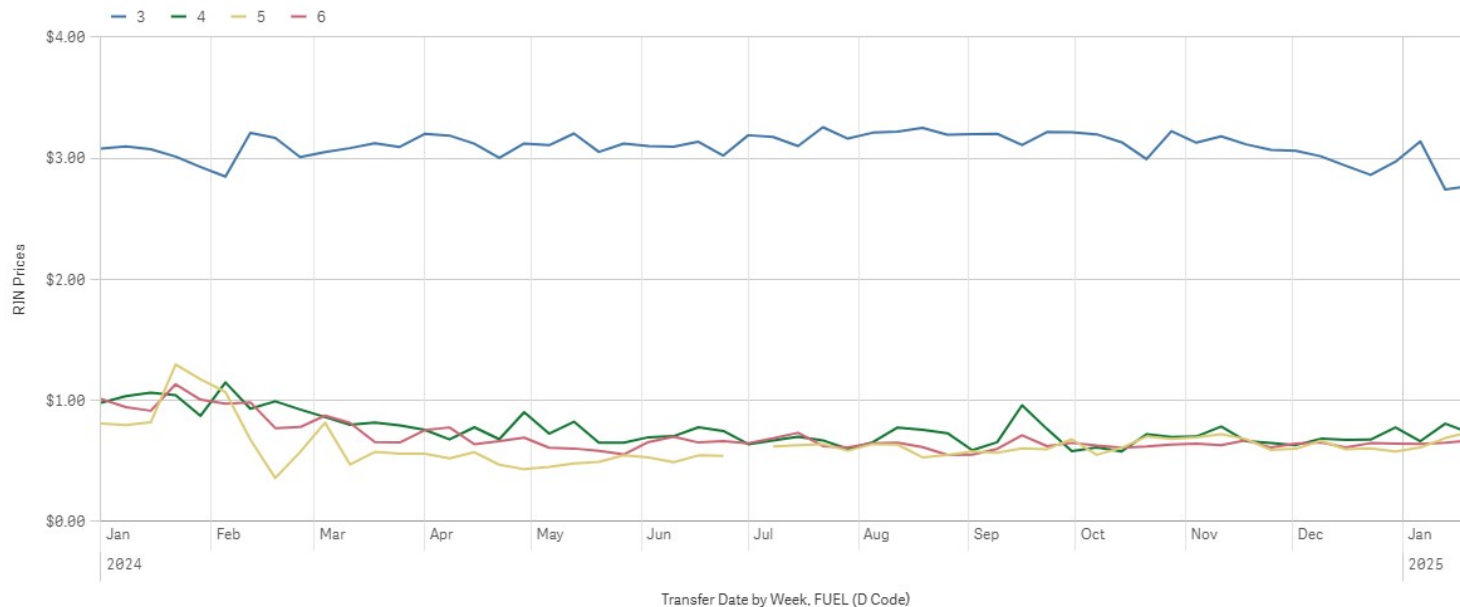


Updated August 2021

This map provides an overview of the major pipelines in the U.S. that move jet fuel. Smaller pipelines and cities supported indirectly by pipeline transport are equally important but are not shown on this map.

Rin Prices

Weekly D3, D4, D5 and D6 RINs Prices



Source: U.S. Environmental Protection Agency (EPA), "RIN Trades and Price Information," retrieved March 15, 2026.

Natural Gas Price

Natural gas spot prices (Henry Hub)

dollars per million British thermal units



Data source: Natural Gas Intelligence

Source: U.S. Energy Information Administration (EIA), Natural Gas Weekly Update.

Waste Pricing

MONTHLY WATER RATES

USAGE TIER	INSIDE CITY RATE	OUTSIDE CITY RATE
Base Charge	\$6.56	\$6.56
>0-3 CCF*	\$2.58 per CCF	\$3.51 per CCF
>3-6 CCF	\$5.34 per CCF	\$6.48 per CCF
>6 CCF	\$6.16 per CCF	\$7.47 per CCF
Irrigation Meter	Same as tiered water rate charges otherwise applicable.	Same as tiered water rate charges otherwise applicable.
Wholesale**	\$3.70	

MONTHLY SEWER RATES

USAGE TIER	RATE
Base Charge	\$6.56
>0-3 CCF*	\$9.74 per CCF
>3-6 CCF	\$13.64 per CCF
>6 CCF	\$15.69 per CCF

Source: City of Atlanta Department of Watershed Management, "Water and Sewer Bill Rates," retrieved March 15, 2026.

Catalyst Costing

	dehydration		oligo-merization 1		2 beds		olefin hydro-generation		Hydro-cracking		olig 2	
bpd/SCFH	1,500		1,500		137,419		1,500		1,500		1,500	
m3/hr	142		19		-		33		137		16	
LHSV/GHSV	1		1		3,986		1		0.5		1	
catalyst	142	m3	19	m3	1,710	lbs	33	m3	273	m3	16	m3
density	680	kg/m3	680	kg/m3	49.6	lb/ft3	680	kg/m3	680	kg/m3	680	kg/m3
loading	96,693	kgs	12,785	kgs	34	ft3	22,406	kgs	185,823	kgs	10,674	kgs
	213,208	lbs	28,190	lbs	776	kgs	49,405	lbs	409,740	lbs	23,536	lbs
lifespan	3	years	3	years	3	years	3	years	3	years	3	years
consumption	71,069	lbs/year	9,397	lbs/year	259	lbs/year	16,468	lbs/year	136,580	lbs/year	7,845	lbs/year
Price	\$10	/lb	\$15	/lb	\$15	/lb	\$20	/lb	\$25	/lb	\$15	/lb
rate	\$710,694	/year	\$140,952	/year	\$3,879	/year	\$329,367	/year	\$3,414,502	/year	\$117,682	/year
cost/load	\$2,132,081		\$422,857		\$11,636		\$988,102		\$10,243,505		\$353,047	